MIT 8103 Advanced Database Systems: Final Summary

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Case study: E-commerce Platform

Overview

The five portfolios examine the design and management of an e-commerce database system that maintains customer, product, order, order-item and payment information. The system supports customer registration, product management, order processing, inventory control and payment records. PostgreSQL was used for relational data management and transaction activities, while MongoDB supported the storage and retrieval of order-related events. The work demonstrates the application of database design, query optimization, transaction management, NoSQL modelling and distributed database concepts.

Portfolio 1: Database Design and Modelling

A relational database comprising five entities was designed: customer, product, order, order_item and payment. Primary and foreign keys establish record identification and relationships between the entities. The order_item table connects orders and products, allowing multiple products within an order while preserving their purchase-time prices.

The PostgreSQL schema incorporates required fields, unique customer emails and CHECK constraints for valid prices, stock and quantities. Sample data comprising two customers, two products, two orders, three order items and one payment was inserted. Constraint tests confirmed that invalid entries, including duplicate emails, missing required values and invalid quantities, were rejected. These measures support data integrity and consistency.

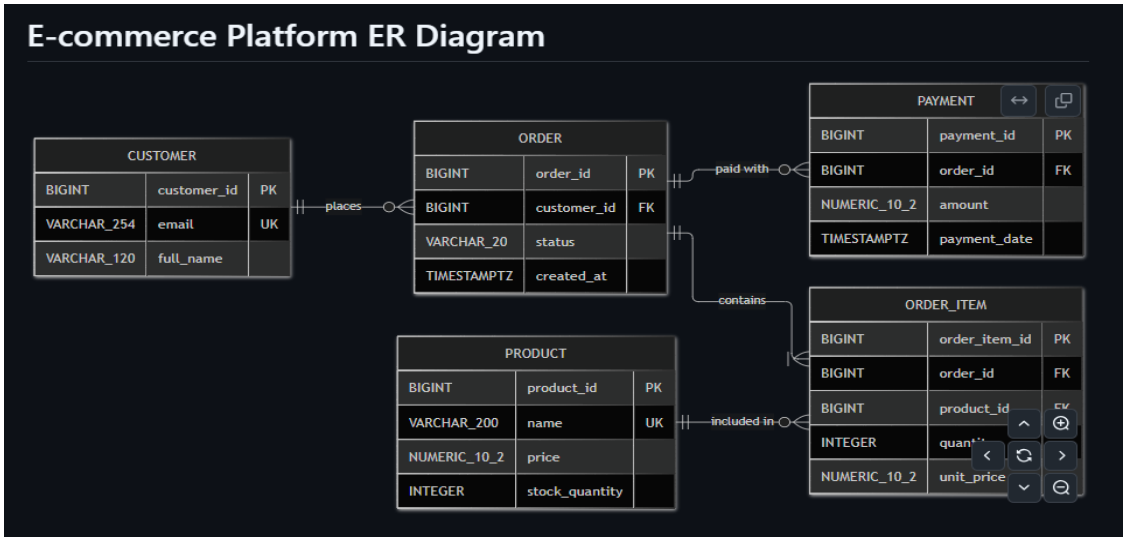


Figure 1. E-commerce database entity-relationship diagram.

Portfolio 2: Query Processing and Optimisation

Three SQL queries retrieved pending orders, available products and customer order counts. Results included two products under King Fisher's pending order, 50 Wireless Mice and 30 Mechanical Keyboards in stock, and one order each for King Fisher and Claire Smith. Index performance was tested using 100,000 synthetic order-history records. A date-range query returned 24 records before and after a B-tree index was created on activity_at.

Measurement	Before Index	After Index
Execution Plan	Sequential Scan	Index Only Scan
Execution times (ms)	5.684, 6.036, 13.189	0.064, 0.056, 0.062
Median (ms)	6.036	0.062
Shared buffer hits	896	3

Table 1. Index performance comparison

The median execution time decreased from 6.036 ms to 0.062 ms, indicating improved performance for the tested query. However, the synthetic dataset and Codespaces environment limit the generalisation of these results. Indexes also require storage and maintenance.

Portfolio 3: Transactions and Concurrency

Transaction testing demonstrated the use of COMMIT and ROLLBACK to preserve valid changes and remove unsuccessful operations. An invalid order-item insertion was rejected by a CHECK constraint, and rollback removed the uncommitted record without affecting an earlier committed update.

A two-session READ COMMITTED test demonstrated row-level locking. Session B waited while Session A held an uncommitted stock update. After Session A committed the reduction of keyboard stock to zero, Session B resumed, but its conditional update affected no rows. The final stock remained zero, demonstrating transaction visibility and the role of conditional updates in preventing overselling.

Portfolio 4: NoSQL and Advanced Data Models

MongoDB was used to create an order_events collection containing four documents representing order placement, payment receipt and shipment activities. Embedded customer, order and event details supported flexible document retrieval. Queries returned four documents, two events for order 101 and one matching payment event. The MongoDB records remained separate from the PostgreSQL database.

Portfolio 5: Distributed and Cloud Database Exercise

A three-member MongoDB replica set was proposed, comprising one primary and two secondary members across separate failure zones. The design considered primary elections, majority write concern, temporary write interruptions and secondary replication delays. Sharding was unnecessary for the four-document demonstration dataset. Authentication, encrypted connections, restricted network access and backup restoration were also identified as deployment considerations. This remained a proposed design; no multi-server deployment or failover test was verified.

Evidence

Practical evidence, including design documents, SQL scripts, MongoDB files, execution results and screenshots, is organised in the five portfolio folders.

Repository: https://github.com/DRaimi/Damilola_Raimi_301910792

Individual Reflection

I found database creation, SQL troubleshooting and MongoDB connectivity challenging. Query optimisation also required careful comparison of execution plans and performance measurements. The concurrency exercise was particularly instructive because coordinating Session A and Session B required understanding transaction locks and waiting for Session A to commit before Session B could complete its update. These challenges taught me the importance of checking and double-checking commands, outputs and transaction results rather than assuming that an operation has produced the intended outcome. Overall, the exercises improved my practical understanding of relational and document databases, transaction behaviour and systematic troubleshooting.

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AI-use Declaration

ChatGPT was used as a learning and support tool to explain database application, assist with troubleshooting and SQL development, and review written documentation. I carried out the practical activities in the repository and database environments utilizing available outputs and screenshots as evidence. AI assistance supported the work but did not replace the practical exercises.